In 1931 and early 1932, experimental physicists competed vigorously with each other to test this description of the nucleus. The method was to try to knock some of Rutherford's postulated neutrons out of atomic nuclei by bombarding the nuclei with high-energy radiation. The competition was won in February 1932 by a member of Rutherford's own experimental team, James Chadwick. Chadwick's bombardment succeeded, neutrons emerged in profusion, and they had just the properties that Rutherford had postulated. The discovery was announced with fanfare by newspapers around the world, and naturally it caught Zwicky's attention.

The neutron arrived on the scene in the same year as Baade and Zwicky were struggling to understand supernovae. This neutron was just what they needed, it seemed to Zwicky. Perhaps, he reasoned, the core of a normal star, with densities of, say, 100 grams per cubic centimeter, could be made to implode until it reached a density like that of an atomic nucleus, 10^{14} (100 trillion) grams per cubic centimeter; and perhaps the matter in that shrunken stellar core would then transform itself into a "gas" of neutrons—a "neutron star" Zwicky called it. If so, Zwicky computed (correctly in this case), the shrunken core's intense gravity would bind it together so tightly that not only would its circumference have been reduced, but so would its mass. The stellar core's mass would now be 10 percent lower than before the implosion. Where would that 10 percent of the core's mass have gone? Into explosive energy, Zwicky reasoned (correctly again; see Figure 5.1 and Box 5.2).

If, as Zwicky believed (correctly), the mass of the shrunken stellar core is about the same as the mass of the Sun, then the 10 percent of it that is converted to explosive energy, when the core becomes a neutron star, would produce 10^{46} joules, which is close to the energy that Zwicky thought was needed to power a supernova. The explosive energy might heat the outer layers of the star to an enormous temperature and blast them off into interstellar space (Figure 5.1), and as the star exploded, its high temperature might make it shine brightly in just the manner of the supernovae that he and Baade had identified.